Lecture 2: Practice Problems

Linear Algebra

November 10, 2026

1 Norms

Problem 1.1 (Mandatory). (a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$ for $\mathbf{x} = (2, -1, 2, -4)^\top$.

(b) Compute $\|\mathbf{y}\|_p$ for $\mathbf{y} = (3, 4)^\top$ and $p = 1, 2, 4, \infty$. What do you observe as p grows?

(c) Show that $\|\mathbf{x}\|_\infty \le \|\mathbf{x}\|_p \le n^{1/p}\|\mathbf{x}\|_\infty$ for every $\mathbf{x} \in \mathbb{R}^n$ and $p \ge 1$, and conclude that $\|\mathbf{x}\|_p \to \|\mathbf{x}\|_\infty$ as $p \to \infty$.

Problem 1.2 (Mandatory). Which of the following functions are norms on $\mathbb{R}^2$? For each norm, sketch its unit ball $\{\mathbf{x} : \|\mathbf{x}\| \le 1\}$; for each function that is not a norm, find an axiom that fails.

(a) $f(\mathbf{x}) = |x_1| + 2|x_2|$;

(b) $f(\mathbf{x}) = |x_1| + x_2^2$;

(c) $f(\mathbf{x}) = |x_1 - x_2|$;

(d) $f(\mathbf{x}) = \max\{|x_1|, |x_2|\} + |x_1 + x_2|$;

(e) $f(\mathbf{x}) = \left(\sqrt{|x_1|} + \sqrt{|x_2|}\right)^2$.

2 Inner Products

Problem 2.1 (Mandatory). Which of the following functions are inner products on $\mathbb{R}^2$? For each inner product, find the symmetric positive definite matrix A with $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top A \mathbf{y}$.

(a) $x_1 y_1 + 2x_2 y_2$;

(b) $x_1 y_1 - x_2 y_2$;

(c) $2x_1 y_1 + x_1 y_2 + x_2 y_1 + x_2 y_2$;

(d) $x_1 y_2 + x_2 y_1$;

(e) $x_1 y_1 + x_1 y_2 + 2x_2 y_2$;

(f) $4x_1 y_1 + 2x_1 y_2 + 2x_2 y_1 + x_2 y_2$.

Problem 2.2 (Mandatory). Consider the inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top A \mathbf{y}$ on $\mathbb{R}^2$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

(a) Compute $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle$, $\|\mathbf{e}_1\|$, $\|\mathbf{e}_2\|$ and the angle between $\mathbf{e}_1$ and $\mathbf{e}_2$ with respect to this inner product.

(b) Find a nonzero vector that is orthogonal to $\mathbf{e}_1$ with respect to this inner product.

(c) Apply the Gram–Schmidt process to $\mathbf{e}_1, \mathbf{e}_2$ with respect to this inner product to find an orthonormal basis.

Problem 2.3. (a) For which $t \in \mathbb{R}$ is $\begin{pmatrix} 1 & t \\ t & 4 \end{pmatrix}$ positive definite? And $\begin{pmatrix} t & 1 \\ 1 & t \end{pmatrix}$?

(b) Use Sylvester's criterion to show that $\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ is positive definite. (Compare with the tridiagonal determinants in the practice problems of Lecture 1.)

Problem 2.4 ($\star$)**.** Consider the Frobenius inner product $\langle A, B \rangle = \operatorname{tr}(A^\top B)$ on $\mathbb{R}^{n \times n}$.

(a) Show that every symmetric matrix is orthogonal to every skew-symmetric matrix.

(b) Recall from Lecture 1 that every square matrix can be written as $A = S + K$ with S symmetric and K skew-symmetric. Show that $\|A\|_F^2 = \|S\|_F^2 + \|K\|_F^2$.

3 Lengths and Distances

Problem 3.1 ($\star$)**.** Use the Cauchy–Schwarz inequality to solve the following problems.

(a) Find the largest value of $2x + 3y + 6z$ subject to $x^2 + y^2 + z^2 = 1$, and the point where it is attained.

(b) Show that $(a + b + c)(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}) \geq 9$ for all $a, b, c > 0$. When does equality hold?

(c) Show that $(x_1 + 2x_2 + 2x_3)^2 \leq 9(x_1^2 + x_2^2 + x_3^2)$.

Problem 3.2. Let x and y be vectors in an inner product space with $\|x + y\| = 7$ and $\|x - y\| = 5$.

(a) Compute $\langle x, y \rangle$.

(b) If also $\|x\| = 5$, compute $\|y\|$.

Problem 3.3 (**Mandatory**)**.** (a) Compute the distance between $(1, 2, 3)^\top$ and $(4, 0, -3)^\top$ with respect to the ℓ_1, ℓ_2 and ℓ_∞ norms, and with respect to the norm induced by the inner product $\mathbf{x}^\top \operatorname{diag}(1, 4, 1)\, \mathbf{y}$.

(b) Prove the reverse triangle inequality $\big|\|x\| - \|y\|\big| \leq \|x - y\|$ for every norm.

4 Angles and Orthogonality

Problem 4.1 (**Mandatory**, $\star$)**.** Find the angle between:

(a) $(2, 1, -1)^\top$ and $(1, -1, 1)^\top$;

(b) $(1, 2, 2)^\top$ and $(2, -1, 2)^\top$;

(c) a diagonal of a cube and one of its edges (Hint: $(1,1,1)$ and $(1,0,0)$);

(d) two diagonals of faces of a cube that start at the same vertex.

Problem 4.2. Find all vectors in $\mathbb{R}^3$ that are orthogonal to both $(1,2,-1)^\top$ and $(2,-1,3)^\top$.

Problem 4.3 ($\star$)**.** Let $Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$.

(a) Show that Q is orthogonal and symmetric, and conclude that $Q^2 = I_3$.

(b) Compute $Q\mathbf{x}$ for $\mathbf{x} = (1,1,1)^\top$, and check that $\|Q\mathbf{x}\| = \|\mathbf{x}\|$.

(c) Show that $Q = I_3 - 2\mathbf{u}\mathbf{u}^\top$ for the unit vector $\mathbf{u} = \frac{1}{\sqrt{3}}(1,-1,-1)^\top$. Describe Q geometrically, and find $\det(Q)$.

Problem 4.4. Three documents are represented by the counts of four words: $d_1 = (2,0,1,1)^\top$, $d_2 = (6,0,3,3)^\top$ and $d_3 = (0,3,0,1)^\top$.

(a) Compute the cosine similarities of d_1 with d_2 and with d_3.

(b) Compute the Euclidean distances of d_1 to d_2 and to d_3.

(c) Which document is "most similar" to d_1 according to each measure? Which answer makes more sense, and why?

5 Orthonormal Bases

Problem 5.1 (**Mandatory**, $\star$)**.** (a) Apply the Gram–Schmidt process to $\mathbf{v}_1 = (1,1,1,1)^\top$, $\mathbf{v}_2 = (1,1,0,0)^\top$ and $\mathbf{v}_3 = (1,0,0,0)^\top$.

(b) Find a fourth vector that extends the result to an orthonormal basis of $\mathbb{R}^4$. (The resulting basis is the Haar basis, which is the starting point of wavelet methods in signal and image processing.)

Problem 5.2 (**Mandatory**)**.** Let $b_1 = \frac{1}{3}(1,2,2)^\top$, $b_2 = \frac{1}{3}(2,1,-2)^\top$ and $b_3 = \frac{1}{3}(2,-2,1)^\top$.

(a) Show that b_1, b_2, b_3 is an orthonormal basis of $\mathbb{R}^3$.

(b) Write $\mathbf{v} = (3,0,3)^\top$ in this basis, without solving a system of equations, and check Parseval's identity.

Problem 5.3. Find the QR decomposition of $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$ with the Gram–Schmidt process.

6 Orthogonal Complements

Problem 6.1 (Mandatory)**.** Let $U = \operatorname{span}\{(1,2,0,1)^\top, (0,1,1,0)^\top\}$. Find a basis of $U^\perp$, and check that $\dim U + \dim U^\perp = 4$.

Problem 6.2. Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$.

(a) Find bases of the four subspaces $\operatorname{col}(A)$, $\operatorname{null}(A^\top)$, $\operatorname{row}(A)$ and $\operatorname{null}(A)$, and check that $\operatorname{null}(A) = \operatorname{row}(A)^\perp$ and $\operatorname{null}(A^\top) = \operatorname{col}(A)^\perp$.

(b) For which $\mathbf{b} \in \mathbb{R}^3$ does the system $A\mathbf{x} = \mathbf{b}$ have a solution? Express your answer as an orthogonality condition.

Problem 6.3 (Mandatory)**.** A linear classifier on $\mathbb{R}^2$ has the weight vector $\mathbf{w} = (2,-1)^\top$ and the bias $b = 1$.

(a) Sketch the decision boundary and the normal vector $\mathbf{w}$.

(b) Classify the points $(1,1)$, $(-1,0)$ and $(0,2)$, and compute their distances to the decision boundary. Which point is classified with the largest margin?

Problem 6.4 ($\star$)**.** Find an equation $\mathbf{w}^\top\mathbf{x} = c$ of the plane through the points $(1,0,0)$, $(0,2,0)$ and $(0,0,3)$, a normal vector of this plane, and the distance from the origin to the plane.

7 Inner Products of Functions

Problem 7.1 (Mandatory, $\star$)**.** Consider $C[0,1]$ with the inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$.

(a) Compute $\langle 1, x\rangle$, $\|1\|$ and $\|x\|$, and the angle between the functions 1 and x.

(b) Apply the Gram–Schmidt process (without normalizing) to $1, x$.

(c) Find the polynomial p of degree at most 1 that minimizes $\int_0^1 (x^2 - p(x))^2\,dx$.

Problem 7.2. Show that, on $C[-\pi,\pi]$, the functions $\sin(mx)$ and $\sin(nx)$ are orthogonal for positive integers $m \neq n$, that $\sin(mx)$ and $\cos(nx)$ are orthogonal for all m, n, and that $\|\sin(nx)\|^2 = \pi$. (Hint: $\sin a \sin b = \frac{1}{2}(\cos(a-b) - \cos(a+b))$.)

8 Orthogonal Projections

Problem 8.1 (Mandatory)**.** Project $\mathbf{x} = (2,1,3)^\top$ onto the line spanned by $\mathbf{b} = (1,1,1)^\top$. Write down the projection matrix, and check that $\mathbf{x} - \pi(\mathbf{x})$ is orthogonal to $\mathbf{b}$.

Problem 8.2 (Mandatory)**.** Let $U = \operatorname{span}\{(1,0,1)^\top, (1,1,0)^\top\}$ and $\mathbf{x} = (1,2,3)^\top$.

(a) Compute $\pi_U(\mathbf{x})$ with the normal equations, and the distance from $\mathbf{x}$ to U.

(b) Find a vector $\mathbf{n}$ with $U^\perp = \operatorname{span}\{\mathbf{n}\}$, and compute $\pi_U(\mathbf{x})$ a second way, as $\mathbf{x} - \pi_{\operatorname{span}\{\mathbf{n}\}}(\mathbf{x})$.

Problem 8.3 ($\star$)**.** Let $\mathbf{1} = (1,\dots,1)^\top \in \mathbb{R}^n$, and let $C = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

(a) Show that C is symmetric and that $C^2 = C$, and that C is the matrix of the orthogonal projection onto $\mathrm{span}\{\mathbf{1}\}^\perp$.

(b) Show that $C\mathbf{x}$ is obtained from $\mathbf{x}$ by subtracting the mean of its entries from every entry, and compute $C\mathbf{x}$ for $\mathbf{x} = (1, 2, 3, 6)^\top$. (This is the "centering" of data used before PCA.)

Problem 8.4 ($\star$). (a) Find the distance from the point $(2, -1, 1)$ to the plane $x - 2y + 2z = 3$, and the closest point of the plane.

(b) Find the distance between the parallel planes $x + 2y + 2z = 1$ and $x + 2y + 2z = 10$.

(c) Find the distance from the point $(1, 2, 3)$ to the line through the origin with direction $(1, 1, 1)$.

Problem 8.5 (Mandatory). Find the least squares line $y = w_0 + w_1 x$ for the data points $(1, 1)$, $(2, 2)$, $(3, 2)$ and $(4, 3)$. Compute the residuals and the sum of squared errors, and check that the residual vector is orthogonal to the columns of the design matrix.

Problem 8.6 ($\star$). Let $(x_1, y_1), \ldots, (x_N, y_N)$ be data points.

(a) Show that the constant c that minimizes $\sum_i (y_i - c)^2$ is the mean $\bar{y} = \frac{1}{N}\sum_i y_i$. Interpret this as a projection onto $\mathrm{span}\{\mathbf{1}\}$.

(b) Show that the line $y = wx$ through the origin that fits the data best (in the least squares sense) has the slope $w = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2}$.

9 Rotations

Problem 9.1 (Mandatory). Let $R = \frac{1}{3}\begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix}$.

(a) Show that R is a rotation, i.e., $R^\top R = I_3$ and $\det(R) = 1$.

(b) Find a unit vector $\mathbf{v}$ on the axis of R, i.e., with $R\mathbf{v} = \mathbf{v}$.

(c) Write $R = QR_3(\theta)Q^\top$ as in the lecture, and use $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ to show that $\mathrm{tr}(R) = 1 + 2\cos\theta$. Find the rotation angle θ.

Problem 9.2. With the rotations R_1 and R_3 of the lecture, compute

$$R_3(\tfrac{\pi}{2})R_1(\tfrac{\pi}{2})\mathbf{e}_2 \qquad \text{and} \qquad R_1(\tfrac{\pi}{2})R_3(\tfrac{\pi}{2})\mathbf{e}_2,$$

and explain the result geometrically.

Problem 9.3 ($\star$). Let $C = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$.

(a) Show that C is a rotation, and that $C^3 = I_3$.

(b) Find the axis of C, i.e., a nonzero vector $\mathbf{v}$ with $C\mathbf{v} = \mathbf{v}$.

(c) A rotation of $\mathbb{R}^3$ by the angle θ about some axis has trace $1 + 2\cos\theta$. Use this to find the angle of C, and compare with part (a).

Problem 9.4. Find $\cos\theta$ and $\sin\theta$ such that the Givens rotation $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ maps $(3,4)^\top$ to a vector of the form $(r,0)^\top$ with $r>0$. What is r?

Problem 9.5 ($\star$)**.** The reflection of $\mathbb{R}^2$ in the line through the origin that makes the angle α with the x-axis is $F_\alpha = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}$.

(a) Check that F_α is orthogonal with determinant -1, and that it fixes the vector $(\cos\alpha, \sin\alpha)^\top$.

(b) Show that $F_\alpha F_\beta = R_{2(\alpha-\beta)}$: the composition of two reflections is a rotation by twice the angle between the lines.

10 The Spectral Theorem

Problem 10.1 (**Mandatory**)**.** Find the spectral decompositions $A = Q\Lambda Q^\top = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$ of

$$A_1 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \qquad \text{and} \qquad A_2 = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}.$$

Which of them is positive definite?

Problem 10.2. Find an orthonormal basis of eigenvectors of $A = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$. (Hint: compare with the example in the lecture.)

Problem 10.3 ($\star$)**.** (a) Find the largest and the smallest value of xy on the circle $x^2+y^2=8$. (Hint: $xy = v^\top A v$ for $v=(x,y)^\top$ and a symmetric matrix A.)

(b) Find the largest and the smallest value of $x^2+4xy+y^2$ on the unit circle, and the points where they are attained.

Problem 10.4. Describe the curves

$$5x^2+4xy+2y^2=1 \qquad \text{and} \qquad x^2+4xy+y^2=1$$

in the plane. Find the directions of their axes, and, for the ellipse, the lengths of its semi-axes. (Hint: write the left-hand side as $\mathbf{v}^\top A \mathbf{v}$ and diagonalize A.)

Problem 10.5 ($\star$)**.** Let P be a symmetric $n \times n$ matrix with $P^2 = P$.

(a) Show that the eigenvalues of P are 0 and 1, and that $P = Q\Lambda Q^\top$ where Λ is a diagonal matrix of 0s and 1s.

(b) Show that P is the orthogonal projection onto the eigenspace E_1, and that $\operatorname{tr}(P) = \operatorname{rank}(P)$.

(c) Check part (b) for $P = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$.

Problem 10.6 (**Mandatory**)**.** Consider the path graph with the vertices $1,2,3,4$ and the edges $\{1,2\}$, $\{2,3\}$, $\{3,4\}$, all with weight 1.

(a) Write down its degree matrix D, its weight matrix W and its Laplacian $L = D - W$.

(b) Compute $\mathbf{x}^\top L\mathbf{x}$ for $\mathbf{x} = (1, 2, 4, 7)^\top$ in two ways: by matrix multiplication, and as a sum over the edges.

(c) Check that $q = (1, \sqrt{2} - 1, 1 - \sqrt{2}, -1)^\top$ is an eigenvector of L, and find its eigenvalue. The eigenvalues of L are 0, $2 - \sqrt{2}$, 2 and $2 + \sqrt{2}$; which split of the path does spectral clustering suggest?

Problem 10.7. Let L be the Laplacian of the cycle $1 - 2 - 3 - 4 - 1$ with weights 1.

(a) Show that the eigenvalues of L are 0, 2, 2 and 4, and find orthonormal eigenvectors.

(b) Compute the ratio cut of the splits $\{1, 2\} \mid \{3, 4\}$, $\{1, 3\} \mid \{2, 4\}$ and $\{1\} \mid \{2, 3, 4\}$, and compare them with λ_2. When does equality hold in $\mathrm{RatioCut}(A) \geq \lambda_2$?

Problem 10.8 ($\star$)**.** Let L be the Laplacian of the complete graph on n vertices, in which all pairs of distinct vertices are joined by edges of weight 1.

(a) Show that $L = nI_n - \mathbf{1}\mathbf{1}^\top$, and find its eigenvalues with their multiplicities.

(b) Show that every split of the vertices has the ratio cut n. What does this mean for spectral clustering?

Problem 10.9. A graph on the vertices $1, \ldots, 5$ has the edges $\{1, 2\}$ with weight 2, $\{2, 3\}$ with weight 1, and $\{4, 5\}$ with weight 3. Find a basis of $\mathrm{null}(L)$ without computing any eigenvalues, and give the multiplicity of the eigenvalue 0. Then find the remaining eigenvalues.

11 The Singular Value Decomposition

Problem 11.1. Find an SVD of $A = \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}$. (Careful: singular values are nonnegative, and they are ordered by size.)

Problem 11.2. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$.

(a) Find the singular values of A, and write $A = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^\top$.

(b) Use the SVD to find bases of the four fundamental subspaces of A.

(c) Compute the pseudo-inverse A^+.

Problem 11.3 (**Mandatory**, $\star$)**.** Let $A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}$.

(a) Compute $AA^\top$ and its eigenvalues and eigenvectors, and use them to find the singular values and the left singular vectors of A. (This is easier than starting from the 3×3 matrix $A^\top A$.)

(b) Find the right singular vectors $\mathbf{v}_i = \frac{1}{\sigma_i} A^\top \mathbf{u}_i$, and complete them to an orthonormal basis of $\mathbb{R}^3$.

(c) Find the best rank-1 approximation A_1 of A, and compute $\|A - A_1\|_F$, $\|A\|_2$ and $\|A\|_F$.

Problem 11.4. (a) Show that A and $A^\top$ have the same singular values.

(b) Show that if A is invertible, then the singular values of A^{-1} are $\frac{1}{\sigma_i}$, and $\|A^{-1}\|_2 = \frac{1}{\sigma_n}$.

(c) Show that $|\det(A)| = \sigma_1\sigma_2\cdots\sigma_n$ for every $n \times n$ matrix A.

(d) Show that for a matrix of rank 1, $\|A\|_2 = \|A\|_F$, and check this for $A = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$.

Problem 11.5 (**Mandatory**, $\star$)**.** (a) Carry out PCA for the four data points $(2, 0)$, $(-2, 0)$, $(0, 1)$, $(0, -1)$: find the covariance matrix, the principal components, the proportion of the variance explained by the first principal component, and the average reconstruction error when the data is projected onto the first principal component.

(b) Carry out PCA for the three data points $(1, 2)$, $(3, 4)$, $(5, 6)$ (center them first!). What is special about this data?

(c) What would go wrong in part (b) if you forgot to center the data? (Hint: in which direction does the first right singular vector of the uncentered data matrix point?)